

iPA Complete TCS Preparation Guide

iON Proctored Assessment • 2025 – 2026

[Syllabus](#) | [Exam Pattern](#) | [Practice Questions](#) | [Bonus Details](#)

■ JOINING BONUS STRUCTURE

SCORE BELOW 55%	SCORE 55% – 79%	SCORE 80% AND ABOVE
No Bonus	Joining Bonus*	Max Joining Bonus*
Offer still valid	Amount decided by TCS	Amount decided by TCS

120 min	100 Marks	2 Sections	1 Attempt
Total Duration	Total Marks	MCQ + Coding	No Reattempt

■ TABLE OF CONTENTS

1. Exam Overview & Bonus Details	2
2. Syllabus — Section-wise Topics	2
3. Exam Pattern & Marking Scheme	3
4. Section A: KYT — Know Your TCS (Q1–Q7)	4
5. Section B: BizSkill — Business Ethics (Q1–Q7)	5
6. Section C: UI — HTML & CSS (Q1–Q8)	6
7. Section D: Unix / Linux Commands (Q1–Q7)	7
8. Section E: Java & Python MCQs (Q1–Q10)	8
9. Section F: SQL / MySQL / PL-SQL (Q1–Q8)	10
10. Section G: Coding Problems — Java & Python (Q1–Q5)	11
11. Preparation Tips & Strategy	13

■ EXAM OVERVIEW & SYLLABUS

TCS IPA — iON Proctored Assessment 2025-26

What is TCS IPA? The iON Proctored Assessment (IPA) is a compulsory post-offer exam conducted at authorised TCS iON centres. It determines your **joining bonus** and BU-Ready status. It does **NOT** affect your offer letter. Since 2025 it is mandatory for all new joiners.

Section-wise Syllabus

#	Section	Key Topics
A	KYT (Know Your TCS)	TCS history, founder, revenue, subsidiaries, core values (EPIC), ISO certifications, TCS Xplore, major clients, recent milestones
B	BizSkill (Business Skills)	Professional ethics, email etiquette, workplace behaviour, situational judgement, conflict resolution, teamwork, POSH Act
C	UI (HTML & CSS)	HTML tags (semantic), forms, tables, CSS selectors, box model, flex/grid basics, inline vs block, JavaScript basics
D	Unix / Linux	Basic commands (ls, cd, mkdir, rm, grep, chmod, chown, find), pipe , redirection >, permissions, shell scripting basics
E	Java / Python (MCQs)	OOP concepts, data types, control flow, arrays, strings, exception handling, collections, recursion, complexity
F	SQL / MySQL / PL-SQL	SELECT, WHERE, JOIN (INNER/LEFT/RIGHT), GROUP BY, HAVING, subquery, aggregate functions, stored procedures, triggers
G	Coding (Java / Python)	Pattern printing, string manipulation, array problems, sorting, searching, basic math problems (2 problems, 70 min)

Exam Pattern & Marking Scheme

Section	Type	No. of Qs	Time	Marks	Negative?
KYT	MCQ	~5	50 min (combined)	~5	No
BizSkill	MCQ	~5	—	~5	No
UI (HTML/CSS)	MCQ	~5	—	~10	No
Unix/Linux	MCQ	~5	—	~10	No
Java / Python	MCQ	~10	—	~20	No
SQL	MCQ	~5	—	~10	No
Coding	Code	2 Problems	70 min	~40	No
TOTAL		~37-40	120 min	100	—

■ SECTION A — KYT (Know Your TCS)

TCS Company Knowledge | ~5 Questions

Q1. TCS was founded in which year?

- A) 1966
- B) 1968 ✓**
- C) 1972
- D) 1981

Explanation: Tata Consultancy Services was established in 1968 by Tata Sons.

Q2. What does the acronym 'EPIC' stand for in TCS core values?

- A) Excellence, Passion, Innovation, Commitment
- B) Ethics, Passion, Integrity, Commitment ✓**
- C) Ethics, Pioneering, Innovation, Creativity
- D) Excellence, Pioneering, Integrity, Commitment

Explanation: TCS core values: Ethics, Passion, Integrity, Commitment.

Q3. Which of the following is a subsidiary / brand of TCS?

- A) Infosys BPO
- B) Syntel
- C) TCS iON ✓**
- D) Mphasis

Explanation: TCS iON is TCS's SMB-focused brand for cloud-based services.

Q4. TCS Xplore is primarily designed for:

- A) Lateral hires only
- B) Fresh campus recruits — pre-joining learning platform ✓**
- C) Senior management training
- D) Client onboarding

Explanation: TCS Xplore is a mandatory pre-joining learning programme for freshers.

Q5. Which TCS flagship programme partners with universities to offer integrated degrees?

- A) TCS NINJA
- B) TCS BPS
- C) TCS TalentNext
- D) TCS Academic Interface Programme (AIP) ✓**

Explanation: TCS AIP partners with over 500 colleges to co-create curriculum.

Q6. In which year did TCS become the first Indian IT company to reach a market cap of \$100 billion?

- A) 2014
- B) 2016
- C) 2018 ✓**
- D) 2020

Explanation: TCS crossed the \$100 billion market cap milestone in 2018.

Q7. TCS headquarters is located in:

- A) Bengaluru

B) Hyderabad

C) Chennai

D) Mumbai ✓

Explanation: TCS is headquartered in Mumbai (Nariman Point), Maharashtra.

■ SECTION B — BizSkill (Business Ethics)

Situational Judgement | ~5 Questions

BizSkill questions are situational — there is no single formula. Always choose the option that reflects **professional integrity, collaboration, and TCS values (EPIC)**.

Q1. You notice a colleague copying code from a public GitHub repository and submitting it as original work for a client deliverable. You should:

- A) Ignore it — it is publicly available anyway
- B) Report it immediately to the client
- C) Inform your team lead / manager about the IP concern ✓**
- D) Copy the same code yourself since it is fair use

Explanation: Using open-source code without proper attribution/license check is an IP risk. Escalate internally first.

Q2. A client asks you to share internal TCS project data to help them benchmark against competitors. You should:

- A) Share the data — the client is paying
- B) Politely decline and explain TCS confidentiality policy ✓**
- C) Share anonymised data without asking anyone
- D) Forward the request directly to your manager and share after

Explanation: Sharing internal data violates TCS's confidentiality agreements. Decline and escalate to your manager.

Q3. You receive an email from an unknown sender offering ₹5,000 to complete a survey about your project. The best course of action is:

- A) Complete the survey — it is just information
- B) Forward it to colleagues to see if they know the sender
- C) Delete and report to TCS IT Security team ✓**
- D) Reply asking for more details before deciding

Explanation: This is a classic phishing/social engineering attempt. Report to IT Security.

Q4. During a client meeting you realise you have made a factual error in a report you submitted. You should:

- A) Hope the client does not notice
- B) Immediately acknowledge the error and provide corrected data ✓**
- C) Wait for the client to raise it
- D) Blame it on data provided by another team

Explanation: Transparency and accountability are TCS core values. Own the mistake and correct it.

Q5. A teammate consistently misses deadlines, affecting the whole sprint. The most professional first step is to:

- A) Complain about them to the project manager
- B) Take over their tasks without saying anything
- C) Have a private, respectful conversation to understand the issue ✓**
- D) Send a formal escalation email immediately

Explanation: Address issues at the lowest level first with empathy before escalating.

Q6. You accidentally send a confidential client email to the wrong recipient. You should:

A) Resend the correct email and hope it is not noticed

B) Immediately inform your manager, recall the email, and follow the data breach process ✓

C) Delete the sent email and say nothing

D) Call the wrong recipient and ask them to ignore it

Explanation: A data breach — however accidental — must be reported immediately per TCS protocol.

Q7. A female colleague reports that she feels uncomfortable with a co-worker's repeated comments. Under the POSH Act you should:

A) Advise her not to take it personally

B) Direct her to the Internal Complaints Committee (ICC) ✓

C) Talk to the co-worker yourself to resolve it informally

D) Ignore it — it is their personal matter

Explanation: Under the POSH Act, complaints must go through the Internal Complaints Committee.

■ SECTION C — UI (HTML & CSS)

Web Basics | ~5–8 Questions

Q1. Which HTML tag is used to create a hyperlink?

- A) <link>
- B) <a> ✓**
- C) <href>
- D) <url>

Explanation: The anchor tag `<a href='...'>` creates hyperlinks in HTML.

Q2. What does the CSS property 'box-model' include? (Select the complete set)

- A) Margin, Border, Padding, Content ✓**
- B) Margin, Shadow, Border, Content
- C) Padding, Border, Outline, Content
- D) Margin, Padding, Font, Content

Explanation: The CSS Box Model layers: Content → Padding → Border → Margin.

Q3. Which CSS selector has the highest specificity?

- A) `p { color: red; }`
- B) `.myClass { color: red; }`
- C) `#myId { color: red; }` ✓**
- D) `* { color: red; }`

Explanation: ID selectors (`#id`) have higher specificity than class (`.class`) or element (tag) selectors.

Q4. What is the correct HTML5 tag to define navigation links?

- A) <nav> ✓**
- B) <navigation>
- C) <menu>
- D) <header>

Explanation: `<nav>` is the semantic HTML5 tag for navigation menus.

Q5. Which CSS display value makes elements appear side-by-side and allows width/height?

- A) inline
- B) block
- C) inline-block ✓**
- D) flex

Explanation: inline-block respects width/height while flowing inline with other elements.

Q6. In CSS Flexbox, which property aligns items along the cross-axis?

- A) justify-content
- B) align-items ✓**
- C) flex-direction
- D) flex-wrap

Explanation: align-items controls alignment on the cross-axis (perpendicular to main axis).

Q7. What will the following HTML render? `<p style='color:red;'>Hello</p>`

- A) Text 'Hello' in the default black colour
- B) Text 'Hello' displayed in red colour ✓**

- C) An error — inline style is invalid
- D) Text 'Hello' with a red background

Explanation: Inline styles directly apply CSS properties to individual elements.

Q8. Which input type in HTML5 shows a date picker in modern browsers?

- A) `<input type="text">`
- B) `<input type="date">` ✓**
- C) `<input type="calendar">`
- D) `<input type="datetime">`

Explanation: `type="date"` renders a native date-picker widget in HTML5-compliant browsers.

■ SECTION D — Unix / Linux Commands

OS Basics | ~5–7 Questions

Q1. Which command lists all files including hidden files in a directory?

- A) ls
- B) ls -l
- C) ls -a ✓**
- D) ls -h

Explanation: `ls -a` shows all files including those starting with a dot (hidden files).

Q2. What does the command 'chmod 755 file.sh' do?

- A) Gives full permission to everyone
- B) Owner: rwx, Group: r-x, Others: r-x ✓**
- C) Owner: rw-, Group: r--, Others: r--
- D) Makes the file read-only for everyone

Explanation: `7=rwx` (owner), `5=r-x` (group), `5=r-x` (others). Standard for executable scripts.

Q3. Which command is used to search for a pattern in a file?

- A) find
- B) locate
- C) grep ✓**
- D) awk

Explanation: `grep` (Global Regular Expression Print) searches for text patterns in files.

Q4. What does the pipe symbol | do in Unix?

- A) Redirects output to a file
- B) Passes the output of one command as input to another ✓**
- C) Runs two commands simultaneously
- D) Appends output to a file

Explanation: The pipe `|` chains commands: `ls -l | grep .txt` sends `ls` output to `grep`.

Q5. Which command shows the current working directory?

- A) cwd
- B) dir
- C) pwd ✓**
- D) cd .

Explanation: `pwd` (Print Working Directory) displays the absolute path of the current directory.

Q6. How do you redirect command output to a file (overwrite)?

- A) command >> file
- B) command > file ✓**
- C) command | file
- D) command < file

Explanation: `>` redirects and overwrites. `>>` appends. `<` reads from file as input.

Q7. Which command removes a directory and all its contents recursively?

- A) rm dir
- B) rmdir -a dir

C) rm -rf dir ✓

D) del -r dir

Explanation: rm -rf: -r for recursive, -f to force without prompting. Use with caution!

■■ SECTION E — Java & Python MCQs

Programming Concepts | ~10 Questions

PYTHON QUESTIONS

Q1. What is the output of: `print(type([]))`?

- A)
- B) ✓**
- C)
- D)

Explanation: `[]` creates an empty list. `type([])` returns `.`

Q2. Which Python data structure does NOT allow duplicate values?

- A) List
- B) Tuple
- C) Set ✓**
- D) Dictionary (values)

Explanation: Sets store only unique elements. Duplicates are automatically removed.

Q3. What is the output of: `print(2 ** 3)`?

- A) 6
- B) 8 ✓**
- C) 9
- D) Error

Explanation: `**` is the exponentiation operator. $2^{**}3 = 2^3 = 8$.

Q4. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except ✓**
- C) error
- D) handle

Explanation: Python uses `try/except` blocks (not `try/catch` like Java).

Q5. What does `list.append(x)` do?

- A) Inserts `x` at the beginning
- B) Adds `x` at the end of the list ✓**
- C) Returns a new list with `x` added
- D) Sorts the list and adds `x`

Explanation: `append()` adds an element to the end in $O(1)$ time.

JAVA QUESTIONS

Q6. Which access modifier makes a class member accessible only within the same class?

- A) public
- B) protected
- C) default
- D) private ✓**

Explanation: `private` limits visibility to the declaring class only.

Q7. What is the output of: `System.out.println(10 / 3);` in Java?

- A) 3.33
- B) 3 ✓**
- C) 3.0
- D) Compile Error

Explanation: Integer division in Java truncates decimal. $10/3 = 3$.

Q8. Which Java keyword prevents method overriding?

- A) static
- B) abstract
- C) final ✓**
- D) synchronized

Explanation: final methods cannot be overridden by subclasses.

Q9. What is the correct way to declare an array in Java?

- A) `int arr = new int[5];`
- B) `int[] arr = new int[5];` ✓**
- C) `int arr[] = new int(5);`
- D) `array arr = new array(5);`

Explanation: `int[] arr = new int[5];` is the standard Java array declaration syntax.

Q10. In OOP, what is 'Encapsulation'?

- A) Hiding implementation details and exposing only necessary interfaces ✓**
- B) One class inheriting from another
- C) A function calling itself
- D) One method having multiple implementations

Explanation: Encapsulation = bundling data + methods, hiding internal state via private fields.

■ ■ SECTION F — SQL / MySQL / PL-SQL

Database Queries | ~5–8 Questions

Q1. Which SQL clause is used to filter rows AFTER grouping?

- A) WHERE
- B) FILTER
- C) HAVING ✓**
- D) ORDER BY

Explanation: HAVING filters grouped results; WHERE filters before grouping.

Q2. What will the query return? SELECT COUNT(*) FROM employees WHERE salary > 50000;

- A) Sum of salaries above 50000
- B) Number of employees earning above 50000 ✓**
- C) All employees with salary > 50000
- D) Average salary above 50000

Explanation: COUNT(*) returns the count of rows matching the WHERE condition.

Q3. Which JOIN returns all rows from the left table and matching rows from the right?

- A) INNER JOIN
- B) RIGHT JOIN
- C) LEFT JOIN ✓**
- D) FULL OUTER JOIN

Explanation: LEFT JOIN keeps all rows from the left table; unmatched right rows show NULL.

Q4. Which keyword removes duplicate rows from a SELECT result?

- A) UNIQUE
- B) DISTINCT ✓**
- C) DIFFERENT
- D) NODUPLICATE

Explanation: SELECT DISTINCT column_name removes duplicate values from the result set.

Q5. What does this query do? UPDATE employees SET salary = salary * 1.10 WHERE dept = 'IT';

- A) Gives all employees a 10% salary cut
- B) Gives IT department employees a 10% raise ✓**
- C) Sets IT employee salary to 1.10
- D) Multiplies all salaries by 10

Explanation: salary * 1.10 multiplies by 1.1, effectively adding 10% to current salary.

Q6. Which aggregate function returns the highest value?

- A) TOP()
- B) MAX() ✓**
- C) HIGHEST()
- D) CEILING()

Explanation: MAX() returns the maximum value in the specified column.

Q7. What is a PRIMARY KEY?

- A) A key that allows NULL values

B) A foreign key reference

C) A column (or set) that uniquely identifies each row and cannot be NULL ✓

D) An index on any column

Explanation: Primary Key = NOT NULL + UNIQUE. Every table should have one.

Q8. Which command permanently removes all rows from a table but keeps its structure?

A) DROP TABLE

B) DELETE FROM table

C) TRUNCATE TABLE ✓

D) REMOVE TABLE

Explanation: TRUNCATE is faster than DELETE (no row-by-row logging) and keeps the table schema.

■ SECTION G — Coding Problems (Java / Python)

2 Problems | 70 Minutes

You must write a **complete working program** (not just a function). You can choose either Java or Python. The judge accepts partial scoring based on test cases passed. Aim to pass at least 50% of cases for partial marks.

Q1. Reverse a String

Write a program that accepts a single string as input and prints the reversed string. Do not use any built-in reverse() function.

Sample Input:

```
HelloWorld
```

Sample Output:

```
dlrowolleH
```

Hint: Use a loop from len-1 to 0, or two-pointer swap approach.

Python Solution:

```
s = input()
result = ''
for i in range(len(s)-1, -1, -1):
    result += s[i]
print(result)
```

Q2. Count Vowels in a String

Given a string, count and print the number of vowels (a, e, i, o, u — both upper and lower case).

Sample Input:

```
Programming
```

Sample Output:

```
3
```

Hint: Iterate over each character and check if it's in 'aeiouAEIOU'.

Python Solution:

```
s = input()
count = 0
for ch in s:
    if ch in 'aeiouAEIOU':
        count += 1
print(count)
```

Q3. Check if a Number is Prime

Write a program that reads an integer N and prints 'Prime' if it is a prime number, otherwise prints 'Not Prime'.

Sample Input:

Sample Output:

Prime

Hint: Check divisibility from 2 to $\sqrt{N}$. If any divisor found → Not Prime.

Python Solution:

```
import math
n = int(input())
if n < 2:
    print('Not Prime')
else:
    is_prime = True
    for i in range(2, int(math.sqrt(n)) + 1):
        if n % i == 0:
            is_prime = False
            break
    print('Prime' if is_prime else 'Not Prime')
```


Q4. Print the Fibonacci Series

Read an integer N and print the first N numbers of the Fibonacci series separated by spaces.

Sample Input:

```
7
```

Sample Output:

```
0 1 1 2 3 5 8
```

Hint: Start with a=0, b=1. Each next = a+b. Swap a=b, b=next.

Python Solution:

```
n = int(input())
a, b = 0, 1
result = []
for _ in range(n):
    result.append(str(a))
    a, b = b, a + b
print(' '.join(result))
```

Q5. Find the Second Largest Element

Given a list of N integers (N >= 2) on one line separated by spaces, find and print the second largest unique element.

Sample Input:

```
10 20 4 45 99 99
```

Sample Output:

```
45
```

Hint: Sort unique values descending and pick index [1].

Python Solution:

```
nums = list(map(int, input().split()))
unique = sorted(set(nums), reverse=True)
print(unique[1])
```

■ PREPARATION TIPS & STRATEGY

Score 80+ and Maximise Your Joining Bonus

- **Timeline**
10–14 days of 2 hrs/day is sufficient for 80+. Start with KYT + BizSkill (easiest marks).
- **Choose Language Early**
Pick Python OR Java and stick to it for the full coding section. Python is faster to write.
- **SQL is High-Value**
SQL questions are direct. Master SELECT, JOIN, GROUP BY, HAVING — often 10-15 marks.
- **Unix = Quick Wins**
~10 commands cover 90% of exam questions. Memorise: ls, cd, grep, chmod, pwd, find, pipe, redirect.
- **UI (HTML/CSS)**
Focus on: semantic tags, box model, selectors, flexbox basics. Not deep — just fundamentals.
- **Coding Strategy**
Read both problems first. Start with the easier one. Partial marks are awarded per test case passed.
- **KYT — Read TCS Xplore**
Complete your TCS Xplore modules — KYT questions are directly from that content.
- **No Reattempt**
You get ONE shot. Prepare properly. There is no second chance once the exam date is set.

55%+ Joining Bonus

80%+ Max Joining Bonus

BU-Ready Shorter Training

All the best! Aim for 80%+ — the bonus amount is decided by TCS and absolutely worth earning with focused preparation.